be minimal for the United States but is not negligible for other markets—but also the risk that the government implicitly defaults by changing the definition of inflation. The Argentinian government, for example, took direct control of its statistics institute in 2007 and since then the discrepancy between its official inflation figures and those reported by independent economists has been up to 15%. Up to 2013, this doctoring has saved the Argentinian government $2.5 billion in linkers payments.[11] The United States periodically changes its inflation basket. While these changes are geared towards keeping the basket relevant for the typical consumer, the basket can change in a way that makes it less relevant for you. This is *time-varying basis risk*. In some countries, however, governments have redefined inflation to the detriment of all consumers. I consider sovereign credit risk in chapter 14 as part of factor investing.

This is not to say that real bonds are not good investments. I recommend that they be used as part of an overall portfolio, rather than constituting the majority of individuals' retirement savings. And whether and how much an investor should hold must be a factor decision, not a decision based on the false notion that real bonds are "real," which I now show.

4.2. REAL BONDS ARE LOUSY INFLATION HEDGES

Almost all investors investing in TIPS hold a portfolio of TIPS, and the composition of that portfolio changes over time (unlike a perfectly immunized bond portfolio). Investors often hold a fixed proportion of their portfolio in TIPS, treating TIPS as an asset class, or as part of an overall bond portfolio. Investing in TIPS this way, as opposed to using individual TIPS bonds to immunize (usually partially unknown) future cash flows, results in TIPS being a poor inflation hedge. Real bonds, unfortunately, are not so real.

Table 11.4 reports means and standard deviations of TIPS and nominal Treasuries from March 1997 to December 2011. The bond portfolios are the BarCap U.S. Treasury and U.S. TIPS benchmarks. The full sample statistics are listed in the top panel of Table 11.4. TIPS show a mean return of 7.0%, exceeding the average Treasury return of 6.2%. TIPS volatility of 5.9% is also higher than the Treasury volatility of 4.7%. TIPS and Treasuries tend to move together, with a correlation of 64%, and the correlations of both TIPS and Treasuries with inflation is low. Far from offering good inflation protection, TIPS' correlation with inflation is just 10%.

The first few years of the TIPS market during the late 1990s and early 2000s were characterized by pronounced illiquidity (as I detail below), so the second panel of Table 11.4 takes data from when the TIPS market was mature and starts at July 2007. In this sample the same stylized facts hold: TIPS have higher average returns than Treasuries (8.4% vs. 7.2%, respectively), TIPS and Treasuries have a relatively high correlation of 45%, and TIPS are poor at hedging inflation with

[11] See *Economist*, "The IMF and Argentina: Motion of Censure," Feb. 9, 2013.